ASHISH KUMAR PARIDA
IL, USA | ashishparida48@gmail.com | +1 872 310 9208

EDUCATION:

Master of Science in Business Analytics **Sept 2023 – June 2025**
DePaul University, Chicago
Bachelor of Technology in Mechanical Engineering **Aug 2019 – Aug 2023**
Institute of Technical Education and Research

PROFESSIONAL EXPERIENCE:

REPHIN INC. **May 2026 – Present**
Data Analytics Lead

- Built a large-scale geospatial data pipeline that ingested dozens of satellite and earth-observation data sources, handling differing resolutions, projections, and coordinate systems, and standardized them into a single clean, analysis-ready dataset covering 500,000+ labeled locations across the western United States.
- Developed automated data-cleaning and preprocessing routines to handle missing, noisy, and misaligned remote-sensing measurements at scale, including spatial alignment, resampling, and validation checks that flagged corrupted data before it reached the model.
- Engineered a consolidated feature set per location, combining spectral, geophysical, and geochemical signals into structured inputs for modeling, and built the indexing logic to query across a continental-scale coordinate grid efficiently.
- Trained a geospatial foundation model from scratch using self-supervised pretraining on large volumes of unlabeled imagery, then fine-tuned it for the downstream task of predicting mineral prospectivity.
- Designed and ran a blind-validation methodology: with deposit locations held out, the model independently rediscovered known major deposits and reached AUC up to 0.97 against ground truth, a rigorous test of real predictive performance rather than in-sample fit.
- Designed and ran a blind-validation methodology: with deposit locations held out, the model independently rediscovered known major deposits and reached AUC up to 0.97 against ground truth, a rigorous test of real predictive performance rather than in-sample fit.
- Lead the company end-to-end, owning technical strategy, data infrastructure decisions, and delivery timelines for geospatial intelligence projects.
- Defined the model architecture and validation roadmap from the ground up, prioritizing engineering effort based on downstream predictive impact.
- Translate complex geospatial modeling results into clear, business-relevant findings for prospective clients and partners.
- Manage the full analytics lifecycle — from raw data ingestion through model deployment — using scalable, reusable Python-based pipelines.

HELIOS HEALTHCARE ADVISORS **Sept 2025 – April 2026**
Data Analyst

- Built and tuned 34 machine learning models across four risk domains (survey deficiencies, federal penalties, star-rating movement, and ownership risk), surfacing predictive signals analysts couldn't see manually.
- Architected HeliosIQ, an end-to-end predictive analytics platform spanning 15,440 skilled nursing facilities nationwide, transforming fragmented CMS data into facility-level risk intelligence that powered the firm's advisory engagements.
- Designed the analytics layer for reuse across 19 quarters and 25.4M records, so each new reporting cycle was a refresh rather than a rebuild, turning a recurring manual effort into infrastructure.
- Applied SHAP-based explainability techniques to translate model outputs into clear, defensible risk narratives for advisory consultants and non-technical stakeholders.
- Built automated reporting pipelines that generated facility-level risk summaries each quarter, reducing manual report preparation time and standardizing deliverables across engagements.
- Designed and maintained PostgreSQL schemas for large-scale CMS datasets, enabling fast, reliable data access for downstream modeling and client reporting.
- Partnered directly with advisory consultants to translate model findings into actionable risk-mitigation recommendations for skilled nursing facility clients.

Deloitte **Sept 2022 – April 2023**
Data Analyst

- Collect, process, and analyze large datasets to help clients make informed business decisions, identifying trends and generating actionable insightsA Deloitte Data Analyst is responsible for collecting, processing, and analyzing large sets of data to help clients make informed business decisions. They use various analytical tools and techniques to identify trends, generate insights, and present actionable recommendations. Deloitte

- Build and maintain dashboards and reports using visualization tools like Power BI, Tableau, QlikView/QlikSense, and AlteryxData Analytics Analyst is responsible for developing data visualization dashboards and advanced reports using any of the leading visualization tools such as Power BI, Tableau, Qlikview, QlikSense, and Alteryx Naukri
- Perform data wrangling, ETL, and data cleansing to prepare raw data for analysis
- Work in Excel for analysis and reporting, applying functions like VLOOKUP/HLOOKUP for day-to-day data tasks
- Work alongside consultants, data engineers, and subject matter experts to deliver solutions tailored to client needs, translating business requirements into analytical tasks, presenting data-driven insights, and participating in team meetings to discuss progress Deloitte
- Support workshops or client presentations, ensuring findings are clearly communicated to both technical and non-technical stakeholders Deloitte
- Communicate regularly with engagement managers, project team members, and representatives from various functional and technical teams, escalating matters that need additional attention ZipRecruiter
- On fraud-focused teams: analyze large datasets to identify trends, patterns, and anomalies related to fraud, and develop/maintain dashboards and reports monitoring suspicious activity, collaborating cross-functionally with legal, operations, and product teams ZipRecruiter
- On digital/marketing analytics teams: leverage advanced analytics and visualization tools to map and optimize end-to-end customer journeys, identifying pain points and opportunities to drive business growth

PROJECT EXPERIENCE:

Fraud & Anomaly Detection Analytics

- Built an end-to-end analytics pipeline to detect fraudulent transactions in financial/payments data, identifying suspicious patterns that rule-based systems miss while keeping false positives low enough to avoid blocking legitimate customers.
- Engineered behavioral and transactional features such as spending velocity, location and time anomalies, and deviation from a customer's normal baseline, to separate genuine activity from fraud signals.
- Developed anomaly-detection and classification models in Python, handling the severe class imbalance inherent in fraud data through targeted sampling and evaluation metrics built for rare-event detection.

Solar Eclipse Weather Balloon Project (NASA-funded), DePaul University

- Analyzed weather-balloon sensor data captured during solar eclipses using Python to process and interpret massive datasets, improving the accuracy of atmospheric models used to predict eclipse behavior by 18%.
- Provided real-time analysis of atmospheric conditions, enabling project managers to make informed decisions that improved mission-strategy success rates by 13%.

SKILLS:

- **Tools and Programming Languages:** Python, SQL, PostgreSQL, pandas, NumPy, PyTorch, scikit-learn, XGBoost, Predictive Modeling, Foundation Model Training, Self-Supervised Learning, Spatial Cross-Validation, SHAP, Automated Reporting, Geospatial/Remote-Sensing Data, CMS/Healthcare Data.
- **Languages:** English, Spanish.
- **Relevant Coursework:** Business Strategy, Game Theory and Strategy.